

# Fractions, Decimals and Percentages

Three ways to write the same number. Convert to compare; recurring decimals become fractions with one algebra trick: multiply, subtract, divide.



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## THE TOPIC ON ONE MAP

### Fractions, Decimals and Percentages

#### 01 Convert and compare

62 out of 80 as a %

65% or  $\frac{5}{8}$ ?

order  $0.2\dot{4}\dot{6}$ ,  $0.24\dot{6}$  ...

#### 02 Recurring decimals: pure

$0.\dot{3}\dot{6} = \frac{4}{11}$

$0.\dot{7}0\dot{2} = \frac{26}{37}$

$4.\dot{5}\dot{7}$

#### 03 Recurring decimals: delayed start

$0.5\dot{7}\dot{2} = \frac{63}{110}$

$0.28\dot{1}\dot{3}$

$0.4\dot{x}$  in terms of  $x$

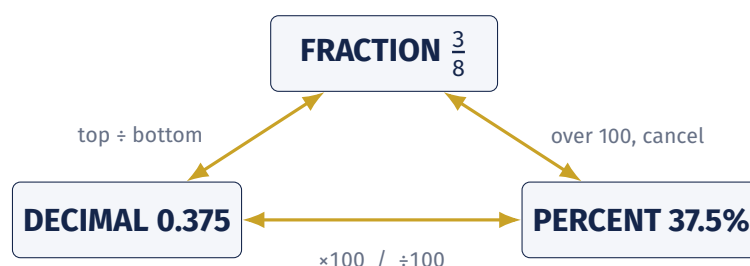
#### 04 Mixed problems

$\frac{3}{4}$  stay, 15% college

Family Railcard

Venn: house and museum

**THE TRIANGLE** Any one form → the other two:



# 01 Convert and Compare

To compare or order, put everything in the same form — usually decimals. For a mark “out of”, divide and multiply by 100.

## WHEN YOU SEE → YOU DO

“62 out of 80”

→  $62 \div 80 \times 100 = 77.5\%$

“did she get a higher percentage in ...”

→ convert both, then compare

“65% of 80 minutes or  $\frac{5}{8}$  of 80 minutes”

→ work out both amounts

“write in order”, recurring dots

→ write each one out to 6 digits

“the lowest mark needed for 60%”

→ 60% of the total, minus what she already has

## WORKED MODEL • ORDER RECURRING DECIMALS

Order  $0.2\dot{4}\dot{6}$ ,  $0.24\dot{6}$ ,  $0.\dot{2}4\dot{6}$ ,  $0.246$ .

1  $0.246464 \dots$ ,  $0.246666 \dots$ ,  $0.246246 \dots$ ,  $0.246$

write them out

ANSWER  $0.246$ ,  $0.\dot{2}4\dot{6}$ ,  $0.2\dot{4}\dot{6}$ ,  $0.24\dot{6}$

IDEA 01

## Practise: Convert and Compare

My questions first, then selected practice.

**Q1** DR ESLAM Compare

CH 1 • TOPIC 8 2 marks

Which is bigger:  $\frac{7}{20}$  or 36%? Show your working.

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## SELECTED PRACTICE

**Q2****PRACTICE**

Practice • Easy

CH 1 • TOPIC 8 2 marks

2 Sasha takes a music exam.

The table shows the result that Sasha can get for different percentages in her music exam.

| Percentage | Result      |
|------------|-------------|
| 50% - 69%  | Pass        |
| 70% - 84%  | Merit       |
| 85% - 100% | Distinction |

Sasha gets 62 out of 80 in her music exam.

What result does Sasha get?

You must show your working.

**(2 marks)**

**Q3****PRACTICE**

Practice • Easy

**CH 1 • TOPIC 8****2 marks**

- 3 Karen got 32 out of 80 in a maths test.  
She got 38% in an English test.

Karen wants to know if she got a higher percentage in maths or in English.  
Did Karen get a higher percentage in maths or in English?

**(2 marks)**

**Q4****PRACTICE**

Practice • Easy

CH 1 • TOPIC 8 4 marks

4 Celina and Zoe both sing in a band.

One evening the band plays for 80 minutes.

Celina sings for 65% of the 80 minutes.

Zoe sings for  $\frac{5}{8}$  of the 80 minutes.

Celina sings for more minutes than Zoe sings.

Work out for how many more minutes.

You must show all your working.

(4 marks)

**Q5****PRACTICE**

Practice • Easy

CH 1 • TOPIC 8 1 mark

12 Circle the largest number.

|              |      |       |                      |
|--------------|------|-------|----------------------|
| 0. $\dot{5}$ | 0.55 | 0.545 | 0.5 $\dot{4}\dot{5}$ |
|--------------|------|-------|----------------------|

(1 mark)

**Q6****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 2 marks

- 2 Write these numbers in order of size.  
Start with the smallest number.

|                     |               |                     |         |
|---------------------|---------------|---------------------|---------|
| $0.2\dot{4}\dot{6}$ | $0.24\dot{6}$ | $0.\dot{2}4\dot{6}$ | $0.246$ |
|---------------------|---------------|---------------------|---------|

(2 marks)

**Q7****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 4 marks

- 11 James is taking three examination papers in Spanish. Here are his first two results.

Paper 1:  $\frac{43}{80}$       Paper 2:  $\frac{38}{65}$

Paper 3 is out of 95.

The marks in each of the three papers are added together.

Find the lowest mark that James needs in Paper 3 to achieve 60% of the total marks.

(4 marks)

## 02 Recurring Decimals: Pure

The repeat starts straight after the point. Multiply by  $10^{(\text{length of the repeat})}$ , subtract, divide.

### WHEN YOU SEE → YOU DO

“use algebra to show that”

→  $x = \dots$ , then  $10x$ ,  $100x$  or  $1000x$

2 digits repeat:  $0.\dot{3}\dot{6}$

→  $100x$

3 digits repeat:  $0.\dot{3}0\dot{6}$

→  $1000x$

a whole number in front:  $4.\dot{5}\dot{7}$

→ do the decimal part, add the 4 back

$x = \text{the decimal}$

Multiply by  $10^n$

Subtract

Divide and cancel

### WORKED MODEL • JUNE 2025 P2H

Show that  $0.\dot{7}0\dot{2} = \frac{26}{37}$ .

1  $x = 0.702702 \dots$

name it

2  $1000x = 702.702702 \dots$

3 digits repeat

3  $999x = 702$

subtract: the tails cancel

ANSWER  $x = \frac{702}{999} = \frac{26}{37}$

## IDEA 02

**Practise: Recurring Decimals: Pure**

My questions first, then selected practice, then real past papers.

**Q8** **DR ESLAM** Pure repeat

CH 1 • TOPIC 8 2 marks

Use algebra to show that  $0.\dot{4}\dot{5} = \frac{5}{11}$ .

**SELECTED PRACTICE****Q9** **PRACTICE** Practice • Medium

CH 1 • TOPIC 8 3 marks

3 Prove that the recurring decimal  $0.\dot{3}\dot{6} = \frac{4}{11}$

(3 marks)

**Q10****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 2 marks

4 Show that the recurring decimal  $0.\dot{3}9\dot{6} = \frac{44}{111}$

(2 marks)

**Q11****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 2 marks

5 Use algebra to show that the recurring decimal  $0.\dot{4}1\dot{7} = \frac{139}{333}$

(2 marks)

**Q12****PRACTICE**

Practice • Medium

**CH 1 • TOPIC 8 5 marks**

12 (a) Without using a calculator, show that  $0.\dot{1}\dot{9}$  can be written as  $\frac{19}{99}$ .

**(3 marks)**

(b) Explain how  $\frac{19}{99} = 0.\dot{1}\dot{9}$  can be used to find  $\frac{19}{990}$  as a decimal and write down its value.

$$\frac{19}{990} = \dots\dots\dots$$

**(2 marks)**

## PAST PAPERS

**Q13****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q15

**CH 1 • TOPIC 8 5 marks**

15 (a) Use algebra to show that  $4.\dot{5}\dot{7} = 4\frac{19}{33}$

(b) Show that  $\frac{2}{6 - 3\sqrt{2}}$  can be written in the form  $\frac{a + \sqrt{a}}{b}$

where  $a$  and  $b$  are integers.  
Show your working clearly.

(Total for Question 15 is 5 marks)

**Q14****PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q18

**CH 1 • TOPIC 8 2 marks**

18 Use algebra to show that  $0.\dot{3}0\dot{6} = \frac{34}{111}$

(Total for Question 18 is 2 marks)

**Q15****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q18

**CH 1 • TOPIC 8 2 marks**

18 Use algebra to show that  $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

## 03 Recurring Decimals: Delayed Start

Some digits come before the repeat. Use **two** multiples of  $x$  whose decimal tails are identical, then subtract them.

### WHEN YOU SEE → YOU DO

$0.5\dot{7}\dot{2}$  (one digit, then a 2-digit repeat)

→  $1000x$  and  $10x \rightarrow 990x$

$0.2\dot{5}$  (one digit, then a 1-digit repeat)

→  $100x$  and  $10x \rightarrow 90x$

$0.281\dot{3}$

→  $10\,000x$  and  $100x \rightarrow 9900x$

“find  $0.4\dot{x}$  in terms of  $x$ ”

→  $100n - 10n = 90n: \frac{36 + x}{90}$

### ZONE RULE

#### line up the tails

write both multiples out; the digits after the point must match exactly

### ZONE CHECK

#### calculator

type the fraction: the display must show your decimal

### WORKED MODEL • JANUARY 2020 P1HR

Show that  $0.5\dot{7}\dot{2} = \frac{63}{110}$ .

1  $1000x = 572.7272 \dots$ ,  $10x = 5.7272 \dots$

same tails

2  $990x = 567$

subtract

ANSWER  $x = \frac{567}{990} = \frac{63}{110}$

× **TRAP** Subtracting  $x$  from  $100x$  when the repeat is delayed: the tails do not match, and nothing cancels.

IDEA 03

## Practise: Recurring Decimals: Delayed Start

My questions first, then selected practice, then real past papers.

**Q16** DR ESLAM Delayed repeat

CH 1 • TOPIC 8 2 marks

Use algebra to show that  $0.1\dot{6} = \frac{1}{6}$ .

### SELECTED PRACTICE

**Q17** PRACTICE Practice • Easy

CH 1 • TOPIC 8 2 marks

5 Prove algebraically that the recurring decimal  $0.2\dot{5}$  has the value  $\frac{23}{90}$

(2 marks)

**Q18****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 2 marks

8 Use algebra to show that the recurring decimal  $0.2\dot{5}\dot{4} = \frac{14}{55}$

(2 marks)

**Q19****PRACTICE**

Practice • Hard

CH 1 • TOPIC 8 2 marks

5 Prove algebraically that the recurring decimal  $0.3\dot{1}\dot{8}$  can be written as  $\frac{7}{22}$

(2 marks)

**Q20****PRACTICE**

Practice · Hard

CH 1 · TOPIC 8 3 marks

- 12 Write  $0.4\dot{1}\dot{6}$  as a fraction in its simplest form.  
You must **show full working** in support of your answer.

(3 marks)

**PAST PAPERS****Q21****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1HR · Q13

CH 1 · TOPIC 8 4 marks

- 13 (a) Use algebra to show that  $0.5\dot{7}\dot{2} = \frac{63}{110}$

Given that  $y$  is a prime number,

- (b) express  $\frac{3}{2 - \sqrt{y}}$  in the form  $\frac{a + b\sqrt{y}}{c - y}$  where  $a$ ,  $b$  and  $c$  are integers.

(Total for Question 13 is 4 marks)

**Q22****PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q16

**CH 1 • TOPIC 8 2 marks**

16 Use algebra to show that the recurring decimal  $0.28\dot{1}\dot{3} = \frac{557}{1980}$

(Total for Question 16 is 2 marks)

**Q23****PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q18

**CH 1 • TOPIC 8 3 marks**

18  $0.4\dot{x}$  is a recurring decimal.

$x$  is a whole number such that  $1 \leq x \leq 9$

Find, in terms of  $x$ , the recurring decimal  $0.4\dot{x}$  as a fraction.

Give your fraction in its simplest form.

Show clear algebraic working.

(Total for Question 18 is 3 marks)

**Q24** **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 2HR • Q17**CH 1 • TOPIC 8** **2 marks**

17 Use algebra to show that  $0.3\dot{4}\dot{5} = \frac{19}{55}$

(Total for Question 17 is 2 marks)

**Q25** **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1HR • Q16**CH 1 • TOPIC 8** **2 marks**

16 Use algebra to show that  $0.1\dot{7}\dot{6} = \frac{35}{198}$

(Total for Question 16 is 2 marks)

**Q26****PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q15

**CH 1 • TOPIC 8 5 marks**

15 (a) Use algebra to show that  $0.3\dot{7}\dot{2} = \frac{41}{110}$

(b) Express  $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$  in the form  $\sqrt{n}$  where  $n$  is an integer.

Show your working clearly.

.....  
(Total for Question 15 is 5 marks)

**Q27****PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q15

**CH 1 • TOPIC 8 2 marks**

15 Use algebra to show that  $0.7\dot{6}\dot{3} = \frac{42}{55}$

(Total for Question 15 is 2 marks)

**Q28****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q16

**CH 1 • TOPIC 8 2 marks**

16 Use algebra to show that the recurring decimal  $0.6\dot{1}\dot{2} = \frac{101}{165}$

(Total for Question 16 is 2 marks)

**Q29****PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q16

**CH 1 • TOPIC 8 2 marks**

16 Use algebra to show that  $0.3\dot{7}\dot{8} = \frac{25}{66}$

(Total for Question 16 is 2 marks)

## 04 Mixed Problems

Fractions, decimals and percentages of one whole in one question: change each part to a number of people (or pounds), then finish the story.

### WHEN YOU SEE → YOU DO

“15% ...  $\frac{3}{4}$  ... the rest”

→ work out each part of the total, subtract from the total

“ $\frac{1}{3}$  off adult tickets, 60% off child tickets”

→ price after each discount

“percentage profit”

→ total income vs total cost

“you may use the Venn diagram”

→ put each group in its region, then multiply by its price

### WORKED MODEL • STUDENTS

240 students: 15% college,  $\frac{3}{4}$  stay at school, the rest do not know. How many do not know?

1  $0.15 \times 240 = 36$ ,  $\frac{3}{4} \times 240 = 180$

each part

ANSWER  $240 - 36 - 180 = 24$

## IDEA 04

**Practise: Mixed Problems**

My questions first, then selected practice.

**Q30** **DR ESLAM** Parts of a whole

CH 1 • TOPIC 8 3 marks

200 people: 30% walk,  $\frac{2}{5}$  take the bus, the rest drive. How many drive?

**SELECTED PRACTICE****Q31** **PRACTICE** Practice · Easy

CH 1 • TOPIC 8 4 marks

- 1 Mr Mason asks 240 Year 11 students what they want to do next year. 15% of the students want to go to college.

$\frac{3}{4}$  of the students want to stay at school.

The rest of the students do not know.

Work out the number of students who do not know.

**(4 marks)**

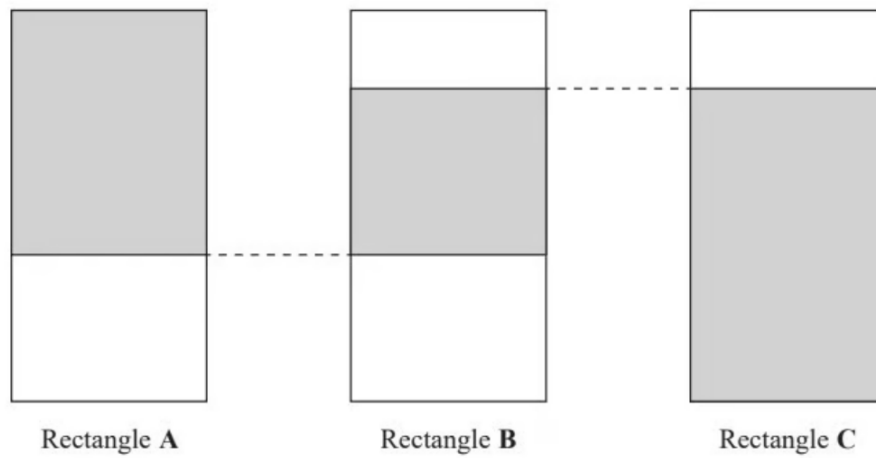
**Q32****PRACTICE**

Practice • Easy

CH 1 • TOPIC 8

**3 marks**

11 The diagram shows three identical rectangles.



$\frac{5}{8}$  of rectangle **A** is shaded.

80% of rectangle **C** is shaded.

What fraction of rectangle **B** is shaded?

**(3 marks)**

**Q33****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8

**4 marks**

- 1 Mr Brown and his 2 children are going to London by train.

An adult ticket costs £24

A child ticket costs £12

Mr Brown has a Family Railcard.

**Family Railcard gives**

$\frac{1}{3}$  off adult tickets

60% off child tickets

Work out the total cost of the tickets when Mr Brown uses his Family Railcard.

**(4 marks)**

**Q34****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 **3 marks**

6 The people working for a company work in Team **A** or in Team **B**.

number of people in Team **A** : number of people in Team **B** = 3 : 4

$\frac{4}{5}$  of Team **A** work full time.

24% of Team **B** work full time.

Work out what fraction of the people working for the company work full time.

Give your fraction in its simplest form.

**(3 marks)**

**Q35****PRACTICE**

Practice • Medium

CH 1 • TOPIC 8

**5 marks**

9 Josh buys and sells books for a living.

He buys 120 books for £4 each.

He sells  $\frac{1}{2}$  of the books for £5 each.

He sells 40% of the books for £7 each.

He sells the rest of the books for £8 each.

Calculate Josh's percentage profit.

**(5 marks)**

Q36

PRACTICE Practice · Hard

CH 1 · TOPIC 8 5 marks

10 At a country park there is a house, a museum and a garden.

The table shows the prices per person to visit the park.

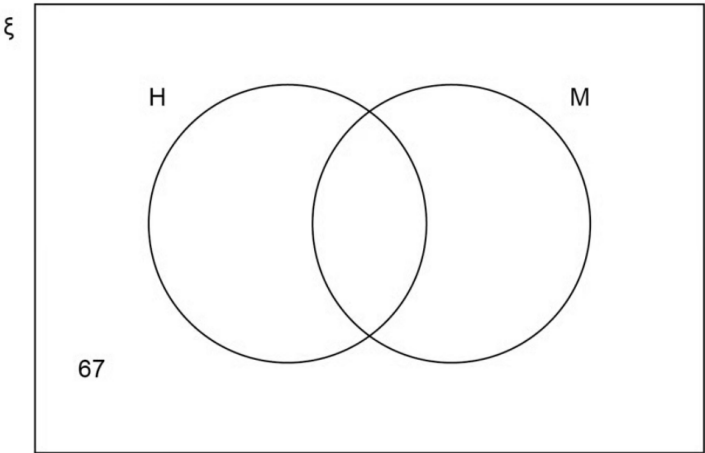
|                  | Price per person |
|------------------|------------------|
| Garden only      | Free             |
| House and museum | £12.50           |
| House only       | £8               |
| Museum only      | £7               |

One day, 480 people visit the park.

- 67 visit the garden **only**.
- 40% visit the house **and** the museum.
- $\frac{3}{8}$  visit the house **only**.

The rest visit the museum **only**.

In total, how much do the 480 people pay to visit the park?  
You may use the Venn diagram to help you.



(5 marks)

# ALL The Whole Topic on One Page

Tear this page out before the exam.

- |           |                       |                                                              |
|-----------|-----------------------|--------------------------------------------------------------|
| <b>01</b> | <b>Convert</b>        | Same form to compare; write recurring decimals out.          |
| <b>02</b> | <b>Pure recurring</b> | $10^n x$ where $n$ = length of the repeat; subtract; divide. |
| <b>03</b> | <b>Delayed</b>        | Two multiples with the same tail; subtract; cancel.          |
| <b>04</b> | <b>Problems</b>       | Each part as an amount, then the story.                      |

**BOTTOM LINE** Recurring decimal → name it  $x$  → two multiples with the same tail → subtract → divide.

# Answers

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Fractions, Decimals and Percentages • for the teacher. Every answer checked with a computer algebra system.

## IDEA 01 • CONVERT AND COMPARE

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- |           |                    |                                                                  |
|-----------|--------------------|------------------------------------------------------------------|
| <b>Q1</b> | Dr Eslam • Compare | $\frac{7}{20} = 35\% < 36\%: 36\%$                               |
| <b>Q2</b> | Practice • Easy    | 77.5%: Merit                                                     |
| <b>Q3</b> | Practice • Easy    | maths 40% > 38%: maths                                           |
| <b>Q4</b> | Practice • Easy    | 52 – 50 = 2 minutes                                              |
| <b>Q5</b> | Practice • Easy    | 0.5                                                              |
| <b>Q6</b> | Practice • Medium  | 0.246, $0.\dot{2}4\dot{6}$ , $0.2\dot{4}\dot{6}$ , $0.24\dot{6}$ |
| <b>Q7</b> | Practice • Medium  | 144 – 81 = 63                                                    |

## IDEA 02 • RECURRING DECIMALS: PURE

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- Q8** Dr Eslam • Pure repeat  $100x - x = 45, x = \frac{45}{99} = \frac{5}{11}$
- Q9** Practice • Medium  $100x - x = 36: \frac{36}{99} = \frac{4}{11}$
- Q10** Practice • Medium  $999x = 396: \frac{44}{111}$
- Q11** Practice • Medium  $999x = 417: \frac{139}{333}$
- Q12** Practice • Medium (a)  $99x = 19$  (b) divide by 10:  $0.0\dot{1}\dot{9}$
- Q13** Edexcel 4MA1 • January 2021 • Paper 1HR • Q15 (a)  $99x = 57, 4 + \frac{57}{99}$  (b)  $\frac{2 + \sqrt{2}}{3}$
- Q14** Edexcel 4MA1 • June 2024 • Paper 2HR • Q18  $999x = 306: \frac{34}{111}$
- Q15** Edexcel 4MA1 • June 2025 • Paper 2H • Q18  $999x = 702: \frac{26}{37}$

### IDEA 03 • RECURRING DECIMALS: DELAYED START

|            |                                               |                                                    |
|------------|-----------------------------------------------|----------------------------------------------------|
| <b>Q16</b> | Dr Eslam • Delayed repeat                     | $100x - 10x = 15, x = \frac{15}{90} = \frac{1}{6}$ |
| <b>Q17</b> | Practice • Easy                               | $100x - 10x = 23: \frac{23}{90}$                   |
| <b>Q18</b> | Practice • Medium                             | $990x = 252: \frac{14}{55}$                        |
| <b>Q19</b> | Practice • Hard                               | $990x = 315: \frac{7}{22}$                         |
| <b>Q20</b> | Practice • Hard                               | $\frac{5}{12}$                                     |
| <b>Q21</b> | Edexcel 4MA1 • January 2020 • Paper 1HR • Q13 | (a) $990x = 567$ (b) $\frac{6 + 3\sqrt{y}}{4 - y}$ |
| <b>Q22</b> | Edexcel 4MA1 • May 2021 • Paper 1H • Q16      | $9900x = 2785: \frac{557}{1980}$                   |
| <b>Q23</b> | Edexcel 4MA1 • November 2021 • Paper 1H • Q18 | $\frac{36 + x}{90}$                                |
| <b>Q24</b> | Edexcel 4MA1 • June 2022 • Paper 2HR • Q17    | $990x = 342: \frac{19}{55}$                        |
| <b>Q25</b> | Edexcel 4MA1 • May 2023 • Paper 1HR • Q16     | $990x = 175: \frac{35}{198}$                       |
| <b>Q26</b> | Edexcel 4MA1 • November 2023 • Paper 2H • Q15 | (a) $990x = 369$ (b) $\sqrt{135}$                  |
| <b>Q27</b> | Edexcel 4MA1 • November 2024 • Paper 2H • Q15 | $990x = 756: \frac{42}{55}$                        |
| <b>Q28</b> | Edexcel 4MA1 • June 2025 • Paper 1HR • Q16    | $990x = 606: \frac{101}{165}$                      |
| <b>Q29</b> | Edexcel 4MA1 • November 2025 • Paper 2H • Q16 | $990x = 375: \frac{25}{66}$                        |

### IDEA 04 • MIXED PROBLEMS

|            |                             |                             |
|------------|-----------------------------|-----------------------------|
| <b>Q30</b> | Dr Eslam • Parts of a whole | $200 - 60 - 80 = 60$        |
| <b>Q31</b> | Practice • Easy             | 24                          |
| <b>Q32</b> | Practice • Easy             | $\frac{17}{40}$             |
| <b>Q33</b> | Practice • Medium           | $16 + 9.60 = \text{£}25.60$ |
| <b>Q34</b> | Practice • Medium           | $\frac{12}{25}$             |
| <b>Q35</b> | Practice • Medium           | 52.5%                       |
| <b>Q36</b> | Practice • Hard             | £4127                       |